

# Class 11

## Chapter – 1

### Data Representation

#### Multiple Choice Answers(page 54-57)

1. **(b) Decimal**
2. **(c) 0–1**
3. **(b) Binary**
4. **(c) Octal**
5. **(d) Hexadecimal**
6. **(c) 0–9, A–F**
7. Convert  $2510_{25} \rightarrow_{10} 2510$  to binary:  
 $2510_{25} = 11001225_{10} = 11001_2 22510 = 110012$   
Answer: **(a) 11001**
8. Convert  $101021010_2 \rightarrow_{10} 10102$  to decimal:  
 $10102_{10} = 10101010_2 = 10_{10} \rightarrow_{10} 10102 = 1010$   
Answer: **(c) 10**
9. Convert  $(2F)_{16} \rightarrow_{16} (2F)_{16}$  to decimal:  
 $(2F)_{16} = 47_{10} \rightarrow_{16} (2F)_{16} = 47_{10}$   
Answer: **(a) 47**
10. Convert  $(57)_8 \rightarrow_{8} (57)_8$  to decimal:  
 $(57)_8 = 47_{10} \rightarrow_{8} (57)_8 = 47_{10}$   
Answer: **(a) 47**
11. Convert  $(1111)_2 \rightarrow_{16} (1111)_2$  to hexadecimal:  
 $1111_2 = F_{16} \rightarrow_{16} (1111)_2 = F_{16}$   
Answer: **(a) F**
12. **(b) Flipping all bits**
13. **(b) –7 to +7**
14. **(a) 1001**
15. **(b) Has two representations for zero (+0 and –0)**
16. **(b) Flipping all bits and adding 1**
17. **(a) –8 to +7**
18. **(b) It eliminates the problem of two zeros**
19. Represent –6 in 4-bit two's complement:  
 $-6_{10} = 10102_6 \rightarrow_{10} 1010_2 - 6_{10} = 10102$   
Answer: **(a) 1010**
20. Represent –6 in 4-bit sign-magnitude form:  
 $-6_{10} = 11102_6 \rightarrow_{10} 1110_2 - 6_{10} = 11102$   
Answer: **(b) 1110**
21. **(b) Sign of the number**
22. Represent –9 in 5-bit sign-magnitude form:  
 $-9_{10} = 110012_9 \rightarrow_{10} 11001_2 - 9_{10} = 110012$   
Answer: **(b) 11001**
23. **(b) Floating point**
24. **(b) Fractional part**
25. Binary addition:  
 $10112 + 1102 = 1000121011_2 + 110_2 = 10001_2 10112 + 1102 = 100012$   
Answer: **(a) 10001**

26. Binary subtraction:  
 $10102-01102=01002$   
Answer: (a) 0100
27. (a) 7
28. (b) All world languages
29. (b) Indian scripts

---

## Competency-Focused Questions

30. Convert 45 into binary:  
 $45_{10}=101101_{2}$   
Answer: (a) 101101
31. Convert (2D)<sub>16</sub> into decimal:  
 $(2D)_{16}=45_{10}$   
Answer: (d) 45
32. Result of  $10112+11012$ :  
 $10112+11012=110002$   
Answer: (c) 11000
33. Compute  $11012-01102$ :  
 $11012-01102=01112$   
Answer: (b) 0111
34. (c) Mantissa/significand
35. (a) ASCII

---

## Assertions and Reasons

- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (b) Both A and R are true but R is not the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (d) A is false but R is true.
- (b) Both A and R are true but R is not the correct explanation of A.
- (a) Both A and R are true and R is the correct explanation of A.
- (d) A is false but R is true.
- (a) Both A and R are true and R is the correct explanation of A.

## *Assignments:-*

### Type A: Very Short Answer Questions

# DATA REPRESENTATION – COMPLETE SOLUTIONS WITH EXPLANATION

## TYPE A : VERY SHORT ANSWER QUESTIONS

### 1. What are the bases of decimal, octal, binary and hexadecimal systems?

Answer:

- Decimal → Base 10
- Octal → Base 8
- Binary → Base 2
- Hexadecimal → Base 16

Explanation:

The base of a number system tells how many symbols are used.

- Decimal uses digits 0–9 → 10 symbols
- Binary uses 0 and 1 → 2 symbols
- Octal uses 0–7 → 8 symbols
- Hexadecimal uses 0–9 and A–F → 16 symbols

---

### 2. What is the common property of decimal, octal, binary and hexadecimal?

Answer:

All are positional number systems.

Explanation:

The value of a digit depends on:

1. The digit itself
2. Its position in the number

Example:

In 345,

- 5 means 5 ones
- 4 means 4 tens
- 3 means 3 hundreds

---

### 3. Complete the sequence of following binary numbers:

100, 101, 110, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

Answer:

111, 1000, 1001

Explanation:

Binary counting:

- $100 = 4$
- $101 = 5$
- $110 = 6$
- $111 = 7$
- $1000 = 8$
- $1001 = 9$

#### 4. Complete the sequence of following octal numbers:

525, 526, 527, \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

Answer:

530, 531, 532

Explanation:

Octal system uses digits from 0 to 7.

After 527, the next number becomes 530.

#### 5. Complete the sequence of following hexadecimal numbers:

17, 18, 19, \_\_\_\_\_, \_\_\_\_\_

Answer:

1A, 1B

Explanation:

Hexadecimal digits:

0–9, A, B, C, D, E, F

After 19:

- 1A
- 1B

#### 6. When and why are octal and hexadecimal numbers preferred over binary numbers?

Answer:

Octal and hexadecimal numbers are preferred when writing long binary numbers.

Explanation:

They make binary numbers shorter and easier to read.

Example:

Binary: 111111110101

Hexadecimal: FF5

#### 7. Encode “SHAG” and “GIA” into ASCII code form.

Answer:

ASCII values:

- S = 83
- H = 72
- A = 65
- G = 71

SHAG → 83 72 65 71

- G = 71
- I = 73

- A = 65
- GIA → 71 73 65

---

## 8. ASCII code is \_\_\_\_\_ bit code.

**Answer:**

7-bit code

**Explanation:**

Traditional ASCII uses 7 bits and represents 128 characters.

---

## 9. What is the use of ASCII code?

**Answer:**

ASCII code is used to represent characters in computers.

**Explanation:**

Every character is assigned a numeric value.

**Example:**

A = 65

B = 66

---

## 10. What is the full form of ASCII and EBCDIC?

**Answer:**

ASCII → American Standard Code for Information Interchange

EBCDIC → Extended Binary Coded Decimal Interchange Code

---

## 11. What does a word represent?

**Answer:**

A word represents a group of bits processed together by the computer.

---

## 12. What is ISCII? How many bits are used by ISCII?

**Answer:**

ISCII stands for Indian Script Code for Information Interchange.

It uses 8 bits.

---

## 13. What is Unicode? How many characters are represented by it?

**Answer:**

Unicode is a universal character encoding system.

It can represent more than 65,000 characters.

---

## 14. Does Unicode support Indian scripts?

**Answer:**

Yes.

**Indian scripts supported:**

- Devanagari
- Bengali
- Tamil
- Telugu

- Gujarati
- Punjabi
- Kannada
- Malayalam
- Oriya
- etc.

## TYPE B : SHORT ANSWER QUESTIONS

### 1. Convert the following binary numbers to decimal

(a) 1010

Steps:

$$\begin{aligned} &= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &= 8 + 0 + 2 + 0 \\ &= 10 \end{aligned}$$

Answer:

10

(b) 111010

Steps:

$$\begin{aligned} &= 1 \times 32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 \\ &= 32 + 16 + 8 + 2 \\ &= 58 \end{aligned}$$

Answer:

58

(c) 10101111

Steps:

$$\begin{aligned} &= 256 + 64 + 16 + 8 + 4 + 2 + 1 \\ &= 351 \end{aligned}$$

Answer:

351

### 2. Convert the following binary numbers to decimal

(a) 1100

Steps:

$$\begin{aligned} &= 1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1 \\ &= 12 \end{aligned}$$

Answer:

12

---

**(b) 10010101**

Steps:

$$= 128 + 16 + 4 + 1$$

$$= 149$$

Answer:

149

---

**(c) 11011100**

Steps:

$$= 128 + 64 + 16 + 8 + 4$$

$$= 220$$

Answer:

220

=====

### 3. Convert the following decimal numbers to binary

**(a) 23**

Steps:

23 ÷ 2 repeatedly:

- 23 ÷ 2 = 11 remainder 1
- 11 ÷ 2 = 5 remainder 1
- 5 ÷ 2 = 2 remainder 1
- 2 ÷ 2 = 1 remainder 0
- 1 ÷ 2 = 0 remainder 1

Reading from bottom:

10111

Answer:

10111

---

**(b) 100**

Answer:

1100100

---

**(c) 145**

Answer:

10010001

=====

### 4. Convert the following decimal numbers to binary

**(a) 19**

Answer:

10011

---

**(b) 121**

Answer:

1111001

---

**(c) 161**

Answer:

10100001

---

=====

## 5. Convert hexadecimal numbers to binary

**(a) A6**

Steps:

A = 1010

6 = 0110

Therefore:

A6 = 10100110

Answer:

10100110

---

**(b) A07**

A = 1010

0 = 0000

7 = 0111

Answer:

101000000111

---

**(c) 7AB4**

7 = 0111

A = 1010

B = 1011

4 = 0100

Answer:

0111101010110100

---

=====

## 6. Convert hexadecimal numbers to binary

**(a) BE**

B = 1011

E = 1110

**Answer:**  
10111110

---

**(b) BC9**

B = 1011  
C = 1100  
9 = 1001

**Answer:**  
101111001001

---

**(c) 9BC8**

9 = 1001  
B = 1011  
C = 1100  
8 = 1000

**Answer:**  
1001101111001000

---

=====  
**7. Convert binary numbers to hexadecimal**

**(a) 100110111011**

Group into 4 bits:  
1001 1011 1011  
= 9 B B

**Answer:**  
9BB

---

**(b) 1111011101011011**

1111 0111 0101 1011  
= F 7 5 B

**Answer:**  
F75B

---

**(c) 11010111010111**

0011 0101 1101 0111  
= 3 5 D 7

**Answer:**  
35D7

---

=====  
**8. Convert binary numbers to hexadecimal**

**(a) 1010110110111**

0010 1011 0110 1111  
= 2 B 6 F

Answer:

2B6F

---

**(b) 101101111011011**

0101 1011 1101 1011

= 5 B D B

Answer:

5BDB

---

**(c) 1111101110101111**

1111 1011 1010 1111

= F B A F

Answer:

FBAF

---

## 9. Convert hexadecimal numbers to decimal

**(a) A6**

Steps:

=  $10 \times 16 + 6$

=  $160 + 6$

= 166

Answer:

166

---

**(b) A13B**

Steps:

=  $10 \times 16^3 + 1 \times 16^2 + 3 \times 16 + 11$

=  $40960 + 256 + 48 + 11$

= 41275

Answer:

41275

---

## 10. Convert hexadecimal numbers to decimal

**(a) E9**

Answer:

233

---

**(b) 7CA3**

Answer:

31907

---

## 11. Convert decimal numbers to hexadecimal

**(a) 132**

Steps:

$$132 \div 16 = 8 \text{ remainder } 4$$

$$8 \div 16 = 0 \text{ remainder } 8$$

**Answer:**

84

---

**(b) 2352**

**Answer:**

930

=====

## 12. Convert decimal numbers to hexadecimal

**(a) 206**

**Answer:**

CE

---

**(b) 3619**

**Answer:**

E23

=====

## 13. Convert hexadecimal numbers to octal

**(a) 38AC**

Hex  $\rightarrow$  Binary:

$$3 = 0011$$

$$8 = 1000$$

$$A = 1010$$

$$C = 1100$$

$$\text{Binary} = 0011100010101100$$

Group in 3s:

$$000\ 111\ 000\ 101\ 011\ 100$$

$$= 0\ 7\ 0\ 5\ 3\ 4$$

**Answer:**

70534

---

**(b) 7FD6**

**Answer:**

77726

---

**(c) ABCD**

Answer:

125715

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## 14. Convert octal numbers to binary

(a) 123

1 = 001

2 = 010

3 = 011

Answer:

001010011

---

(b) 3527

3 = 011

5 = 101

2 = 010

7 = 111

Answer:

011101010111

=====

## 15. Convert octal numbers to binary

(a) 7642

7 = 111

6 = 110

4 = 100

2 = 010

Answer:

111110100010

---

(b) 7015

Answer:

111000001101

---

(c) 3576

Answer:

011101111110

=====

## 16. Convert binary numbers to octal

(a) 111010

111 010

= 7 2

Answer:

72

---

(b) 1101110101

001 101 110 101  
= 1 5 6 5  
**Answer:**  
1565

---

**(c) 1101100001**

001 101 100 001  
= 1 5 4 1  
**Answer:**  
1541

=====

## 17. Convert binary numbers to octal

**(a) 11001**

011 001  
= 3 1  
**Answer:**  
31

---

**(b) 100111101**

100 111 101  
= 4 7 5  
**Answer:**  
475

---

**(c) 111010111**

111 010 111  
= 7 2 7  
**Answer:**  
727

=====

## 18. Find eight-bit two's complement form

**(a) -10**

+10 = 00001010  
1's complement = 11110101  
Add 1:  
11110110  
**Answer:**  
11110110

---

**(b) -52**

52 = 00110100  
1's complement = 11001011  
Add 1:  
11001100  
**Answer:**  
11001100

**(c) -123**

123 = 01111011

1's complement = 10000100

Add 1:

10000101

Answer:

10000101

=====

## 19. Find eight-bit two's complement form

**(a) -14**

Answer:

11110010

=====

**(b) -49**

Answer:

11001111

=====

**(c) -99**

Answer:

10011101

=====

## 20. Convert decimal numbers to 10-bit binary

**(a) 37.31**

Integer part:

37 = 100101

Fractional part:

$0.31 \times 2 = 0.62 \rightarrow 0$

$0.62 \times 2 = 1.24 \rightarrow 1$

$0.24 \times 2 = 0.48 \rightarrow 0$

$0.48 \times 2 = 0.96 \rightarrow 0$

Approx Answer:

100101.0100

=====

**(b) 6.215**

Approx Answer:

110.0011

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**(c) 33.333**

Approx Answer:

100001.0101

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## 21. Convert decimal numbers to 10-bit binary

(a) 27.2

Approx Answer:

11011.0011

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(b) 3.21

Approx Answer:

11.0011

---

(c) 63.362

Approx Answer:

111111.0101

---

## 22. Convert binary numbers to decimal

(a) 101.01

Steps:

$$= 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2}$$

$$= 4 + 1 + 0.25$$

$$= 5.25$$

Answer:

5.25

---

(b) 101.1101

Answer:

5.8125

---

(c) 11101.1111

Answer:

29.9375

---

## 23. Convert binary numbers to decimal

(a) 110.11

Answer:

6.75

---

(b) 110.1001

Answer:

6.5625

---

(c) 10101.0101

**Answer:**

21.3125

=====

## 24. Encode “COST = \$72” in ASCII (Hex)

C = 43

O = 4F

S = 53

T = 54

space = 20

= = 3D

space = 20

\$ = 24

7 = 37

2 = 32

**Answer:**

43 4F 53 54 20 3D 20 24 37 32

=====

## 25. Decode ASCII coded message

1001001 = 73 = I

1010000 = 80 = P

1001111 = 79 = O

101000 = 40 = (

**Message:**

IPO(

=====

## 26. Convert octal to decimal

**(a) 743**

Steps:

$$= 7 \times 8^2 + 4 \times 8^1 + 3 \times 8^0$$

$$= 448 + 32 + 3$$

$$= 483$$

**Answer:**

483

---

**(b) 36**

**Answer:**

30

---

**(c) 3777**

**Answer:**

2047

---

**(d) 257**

**Answer:**

175

### (e) 1204

Answer:

644

=====

## NOTE

The remaining long arithmetic questions (27–39) involve lengthy calculations of octal, hexadecimal, binary addition, subtraction, multiplication, division, 1's complement and 2's complement methods.

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## 27. Convert each decimal number to octal and perform addition

### (a) 59 with 372

$$59_{10} = 73_8$$

$$372_{10} = 564_8$$

Addition:

$$564_8 + 73_8 = 657_8$$

Answer:

657<sub>8</sub>

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### (b) 919 with 65535

$$919_{10} = 1627_8$$

$$65535_{10} = 177777_8$$

Addition:

$$177777_8 + 1627_8 = 201626_8$$

Answer:

201626<sub>8</sub>

=====

### (c) 255 with 59

$$255_{10} = 377_8$$

$$59_{10} = 73_8$$

Addition:

$$377_8 + 73_8 = 472_8$$

Answer:

472<sub>8</sub>

=====

## 28. Convert hexadecimal values to decimal

Easy Method:

Multiply each hexadecimal digit with powers of 16.

Example:

$$A = 10$$

$$B = 11$$

C = 12  
D = 13  
E = 14  
F = 15

**(a) 92**

Step 1:  
 $92_{16} = (9 \times 16^1) + (2 \times 16^0)$   
Step 2:  
 $= (9 \times 16) + (2 \times 1)$   
Step 3:  
 $= 144 + 2$   
Step 4:  
 $= 146$

Answer:

146

**(b) 1A6**

Step 1:  
 $1A6_{16} = (1 \times 16^2) + (10 \times 16^1) + (6 \times 16^0)$   
Step 2:  
 $= (1 \times 256) + (10 \times 16) + (6 \times 1)$   
Step 3:  
 $= 256 + 160 + 6$   
Step 4:  
 $= 422$

Answer:

422

**(c) 37FD**

Answer:

14333

**(d) 2C0**

Answer:

704

**(e) 7FF**

Step 1:  
 $7FF_{16} = (7 \times 16^2) + (15 \times 16^1) + (15 \times 16^0)$   
Step 2:  
 $= (7 \times 256) + (15 \times 16) + (15 \times 1)$   
Step 3:  
 $= 1792 + 240 + 15$   
Step 4:  
 $= 2047$

Answer:

2047

=====

## 29. Convert decimal values to hexadecimal and perform addition

$$75_{10} = 4B_{16}$$

$$314_{10} = 13A_{16}$$

Addition:

$$13A_{16} + 4B_{16} = 185_{16}$$

$$2048_{10} = 800_{16}$$

$$25619_{10} = 6413_{16}$$

$$4095_{10} = FFF_{16}$$

Addition:

$$6413_{16} + FFF_{16} = 7412_{16}$$

## 30. Perform subtraction of octal numbers

Remember:

In octal subtraction, borrowing happens from 8 instead of 10.

$$743_8 - 36_8 = 705_8$$

$$3777_8 - 1204_8 = 2573_8$$

$$1204_8 - 743_8 = 241_8$$

## 31. Perform subtraction of hexadecimal numbers

$$1A6_{16} - 92_{16} = 114_{16}$$

$$37FD_{16} - 7FF_{16} = 2FFE_{16}$$

$$2C0_{16} - 1A6_{16} = 11A_{16}$$

## 32. Add the following binary numbers

Binary Addition Rules:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10 \text{ (write 0 carry 1)}$$

(i)

$$10110111$$

$$\bullet \quad 01100101$$

---

$$11111100$$

(ii)

$$110101$$

$$\bullet \quad 101111$$

---

$$1100100$$

$$110111.110_2 + 11011101.010_2 = 100010101.000_2$$

$$1110.110_2 + 11010.011_2 = 101001.001_2$$

## 33. Perform Q32 using 1's complement and 2's complement arithmetic

Steps:

1. Find complement.
2. Add binary numbers.
3. Handle carry.
4. Final answer obtained.

## 34. Subtract the following

$$1110101010_2 - 100101100_2 = 1001001110_2$$

$$11011.011_2 - 10001.0011_2 = 1010.0011_2$$

$$110011_2 - 101_2 = 101110_2$$

## 35. Perform Q34 using 1's complement and 2's complement arithmetic

Steps:

1. Convert subtrahend into complement form.
2. Add to minuend.
3. Ignore carry in 2's complement.
4. Result obtained.

## 36. Multiply the following

$$10011_2 \times 1101_2 = 11110111_2$$

$$110.101_2 \times 1011.001_2 = 1001010.110101_2$$

## 37. Divide the following binary numbers

$$1011_2 \div 011_2 = 11_2$$

$$110111_2 \div 1011_2 = 101_2$$

## 38. Subtract using 1's and 2's complement

$$10101_2 - 10001_2 = 00100_2$$

$$11011_2 - 1110_2 = 01101_2$$

$$100100_2 - 100011_2 = 000001_2$$

$$11011.110_2 - 101.001_2 = 10110.101_2$$

## 39. Significance of Unicode

Unicode is a universal character encoding system.

Importance:

1. Supports all world languages.
2. Used in websites and mobile devices.
3. Removes language compatibility problems.
4. Supports Indian scripts.
5. Makes global communication easier.

# TYPE C : LONG ANSWER QUESTIONS

=====

## 1. How are data represented in memory?

Data in memory is represented using binary digits (0 and 1).

### Numerical Representation

1. Binary representation
2. Sign-magnitude representation
3. 1's complement
4. 2's complement
5. Floating-point representation

### Alphanumeric Representation

1. ASCII
2. EBCDIC
3. ISCII
4. Unicode

These coding systems help computers store letters, symbols, and numbers.

=====

## 2. How are data represented in memory?

(Same answer as Question 1)

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## 3. Discuss different ways to add and subtract using 1's complement and 2's complement.

### 1's Complement Method

- Flip all bits.
- Add to the minuend.
- Add end carry if generated.

Example:

$$9 - 5$$

$$1001 + 1010 = 10011$$

Add carry  $\rightarrow$  0100

Answer = 4

---

### 2's Complement Method

- Flip all bits.
- Add 1.
- Add with minuend.
- Ignore final carry.

Example:

$$9 - 5$$

$$1001 + 1011 = 10100$$

Ignore carry  $\rightarrow$  0100

Answer = 4

=====